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## Radiation treatment system and method of operating radiation treatment system


### Abstract

To provide a radiation treatment system which enables wide irradiation range of radiation to a patient without increasing a load on a structure body. A radiation treatment system includes: a couch that carries a treatment target; a radiation source; a rotation mechanism configured to support the radiation source and to rotate the radiation source around the couch; a sensor configured to detect radiation transmitted through the treatment target; and a control unit configured to control the radiation source and the rotation mechanism, and the control unit sets an irradiation plan in which an irradiation range of first irradiation and an irradiation range of second irradiation are partially overlapped, and controls a radiation dose for an overlapping portion based on a detection result obtained by the sensor.



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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION
(1) The present application claims priority from Japanese application JP2021-159910, filed on Sep. 29, 2021, the contents of which is hereby incorporated by reference into this application.
BACKGROUND OF THE INVENTION
1. Field of the Invention
(2) The present invention relates to a radiation treatment system and a method of operating a radiation treatment system.
2. Description of the Related Art
(3) A radiation treatment method includes a method of using a linear accelerator called a linac as a radiation source, and emitting radiation to an affected part of a patient, which is previously imaged with an X-ray CT apparatus, an MRI apparatus, or the like, from various directions in three dimensions.
(4) For example, an intensity modulated radiation treatment (IMRT) method of shielding a part of a radiation beam emitted from the radiation source by using a multi-leaf collimator (MLC) and

irradiating the affected part while modulating a radiation intensity distribution of a region irradiated with the radiation, and a volumetric modulated arc therapy (VMAT) in which the multi-leaf collimator and the radiation are rotated around the patient at that time are known.
(5) In addition, an image-guided radiotherapy (IGRT) method in which a radiation treatment apparatus equipped with an X-ray imaging apparatus is used to take an X-ray image immediately before the irradiation with the radiation, and the image is compared with the image taken in advance, thus adjusting an X-ray irradiation position that is determined based on the image taken in advance is also known.
(6) A moving object tracking irradiation method is also known in which when the affected part moves due to a respiratory motion, or the like, a position of the affected part is obtained based on the image taken by the X-ray imaging apparatus, a direction of the radiation source is tilted by a gimbal mechanism, and a radiation irradiation direction is made to follow the position of the affected part.
(7) A tomotherapy method is also known in which the volumetric modulated arc therapy (rotation IMRT) and the image-guided radiotherapy (IGRT) are combined, and further helical irradiation is performed while moving a couch on which the patient is placed.
(8) Also, in order to widen a radiation irradiation range on the patient, JP-A-2008-173182 proposes a method of sequentially performing irradiation with fan beams in adjacent irradiation ranges by tilting a radiation axis from a radiation source at the same position. Further, JP-A-2015-100455 proposes a method of widening the radiation irradiation range by moving an irradiation region by changing a direction of a radiation source while emitting the radiation.
(9) In a radiation treatment, it is required to widen a range in which the irradiation with radiation can be performed and to control an irradiation amount of the radiation with high accuracy.
(10) As in JP-A-2008-173182, when the irradiation is sequentially performed in the adjacent irradiation ranges, the irradiation can be performed in a wide range, but a gap region or an overlapping region is generated between the irradiation ranges, and an excess or deficiency of an irradiation amount of these regions is generated.
(11) In addition, as in JP-A-2015-100455, when the irradiation is performed while changing a direction of an irradiation axis of a radiation apparatus with respect to a rotation axis of the radiation apparatus, a demand for control of the irradiation axis and the MLC at an overlapping region or an irradiation end portion is severe, and it is difficult to control the irradiation amount with high accuracy.
SUMMARY OF THE INVENTION
(12) An object of the invention is to provide a radiation treatment system capable of performing irradiation in a wide range with high accuracy.
(13) In order to achieve the above object, a radiation treatment system of the invention includes: a couch that carries a treatment target; a radiation source; a rotation mechanism configured to support the radiation source and to rotate the radiation source around the couch; a sensor configured to detect radiation transmitted through the treatment target; and a control unit configured to control the radiation source and the rotation mechanism, and the control unit sets an irradiation plan in which an irradiation range of first irradiation and an irradiation range of second irradiation are partially overlapped, and controls a radiation dose for an overlapping portion based on a detection result obtained by the sensor.
(14) According to one method of operating a radiation treatment system of the invention, the radiation treatment system includes: a couch on which a treatment target is placed; a radiation source; a rotation mechanism configured to support the radiation source and to rotate the radiation source around the couch; a sensor configured to detect radiation transmitted through the treatment target; and a control unit configured to control the radiation source and the rotation mechanism, and the method of operating a radiation treatment system executed by the control unit includes: a step of setting an irradiation plan in which an irradiation range of first irradiation and an irradiation

range of second irradiation are partially overlapped, a step of acquiring a detection result by the sensor while executing at least the first irradiation, a step of, based on the detection result, performing correction of a radiation dose emitted to an overlapping portion in subsequent irradiation, and a step of performing irradiation reflecting the correction.
(15) According to the invention, the radiation treatment system enables wide range irradiation with high accuracy.

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## Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a block diagram of a radiation treatment system according to an embodiment of the invention.
(2) FIG. **2** is a perspective view of a radiation treatment apparatus of the embodiment.
(3) FIG. **3** is a diagram showing a schematic configuration of a head swing mechanism in the radiation treatment apparatus of FIG. **2**.
(4) FIG. **4** is a cross-sectional view of a radiation irradiation apparatus of FIG. **3**.
(5) FIG. **5** is a flowchart showing an operation of the radiation treatment system of the embodiment.
(6) FIG. **6** is a flowchart showing an operation of the radiation treatment system of the embodiment.
(7) FIG. **7** is a perspective view showing a configuration example of a leaf group of a multi-leaf collimator of FIG. **4**.
(8) FIGS. **8**A-**1** and **8**A-**2** are diagrams showing first and second irradiation directions and irradiation ranges in the radiation irradiation apparatus of the embodiment, and FIGS. **8**B-**1** and **8**B-**2** are diagrams showing an irradiation direction and an irradiation range of a comparative example.
(9) FIG. **9**A is a diagram showing dose distributions based on irradiation with beams in the first and second irradiation directions of the radiation irradiation apparatus of the embodiment and a total dose distribution thereof, and FIG. **9**B is a diagram showing a dose distribution based on irradiation with beams of the comparative example.
(10) FIG. **10** is a diagram showing the dose distributions based on irradiation with the beams in the first and second irradiation directions of the radiation irradiation apparatus of the embodiment and the total dose distribution thereof.
(11) FIG. **11** is a diagram showing an overlap of the irradiation ranges of the radiation irradiation apparatus of the embodiment.
(12) FIG. **12** is a diagram showing an irradiation range when the first irradiation direction of the radiation irradiation apparatus of the embodiment is shifted in a radial direction.
(13) FIG. **13** is a diagram showing the irradiation range of the comparative example.
(14) FIG. **14** is a diagram showing an irradiation range when the first irradiation direction of the radiation irradiation apparatus of the embodiment is shifted in the radial direction and a rotation axis direction.
(15) FIG. **15** is a diagram showing an irradiation range when the second irradiation direction of the radiation irradiation apparatus of the embodiment is shifted in the radial direction and the rotation axis direction.
(16) FIG. **16** is a diagram showing that when the irradiation direction is shifted in the radial direction to include an isocenter in the irradiation range of the radiation irradiation apparatus of the embodiment, the dose distribution is larger in a vicinity of the isocenter than in surroundings.
(17) FIG. **17** is a diagram of an error of a radiation dose in an overlapping portion of rotation irradiation.

(18) FIG. **18** is a specific example of correction of the radiation dose (No. 1).
(19) FIG. **19** is a specific example of correction of the radiation dose (No. 2).
(20) FIG. **20** is a flowchart showing a processing procedure of the correction of the error.
(21) FIG. **21** is a diagram illustrating irradiation control according to a treatment plan.
(22) FIG. **22** is a diagram illustrating widening of the irradiation range using three times of the rotation irradiation.

DESCRIPTION OF THE PREFERRED EMBODIMENTS

(23) A radiation treatment system according to one embodiment of the invention will be described.
(24) Apparatus Configuration of Radiation treatment System
(25) First, an apparatus configuration of the radiation treatment system of the present embodiment will be described.
(26) As shown in FIG. **1**, a radiation treatment system **10** includes a treatment planning apparatus **11**, a control apparatus (control unit) **12**, and a radiation treatment apparatus **20**.
(27) The treatment planning apparatus **11** receives three-dimensional image data captured in advance for a patient B as a treatment target, and creates, as a treatment plan, properties of radiation to be emitted to the patient B (a dose, a time, an angle, a position, a radiation region, and the like of the radiation to be emitted to the patient B) according to contents of a radiation treatment. Further, in order to emit the radiation with the dose, the time, the angle, and the like of the radiation of the treatment plan, the treatment planning apparatus **11** outputs, to the control apparatus **12**, control parameter values such as a tilt angle of a head swing mechanism **301** of a radiation source to be described later, a rotation angle of a rotation ring **22** with respect to a ring frame **21**, and an irradiation timing of a rotation shaft **25** and a radiation irradiation apparatus **24** as an irradiation plan. A long-term plan can be made as the treatment plan. In the long-term plan, an entire treatment plan is implemented by a combination of a schedule and an individual treatment plan, such as a day of a first time of irradiation treatment and the treatment plan thereof, a day of a second time of irradiation treatment and the treatment plan thereof.
(28) The control apparatus **12** controls an operation of the radiation treatment apparatus **20** based on the control parameter values (irradiation plan) received from the treatment planning apparatus **11**. The control apparatus **12** includes a CPU and a memory, and the CPU reads and executes a program stored in the memory in advance to execute a control operation by software.
(29) FIG. **2** is a perspective view showing a schematic configuration of the radiation treatment apparatus **20**.
(30) As shown in FIG. **2**, the radiation treatment apparatus **20** includes the radiation irradiation apparatus **24**, a rotation mechanism **302**, a couch **28**, and the head swing mechanism **301**. The rotation mechanism **302** supports the radiation irradiation apparatus **24** and rotates the radiation irradiation apparatus **24** around an isocenter C**0**. The couch **28** carries a treatment target site of the patient at the isocenter C**0**. The head swing mechanism **301** is disposed between the radiation irradiation apparatus **24** and the rotation mechanism **302**, and swings the radiation irradiation apparatus **24** to swing an irradiation axis of the radiation emitted from the radiation irradiation apparatus **24**.
(31) The rotation mechanism **302** includes the ring frame **21** and the rotation ring **22**.
(32) The ring frame **21** is disposed such that a rotation central axis C**1** faces a substantially horizontal direction. The rotation ring **22** has a structure in which an outer peripheral surface thereof is supported by an inner peripheral surface of the ring frame **21** and the rotation ring **22** can rotate along the inner peripheral surface of the ring frame **21**. The rotation ring **22** is driven by a rotation drive mechanism (not shown) and revolves around the rotation central axis C**1**.
(33) The rotation shaft **25** extending downward is integrally formed on the outer peripheral surface of a lower end portion **21***a* of the ring frame **21**, and the rotation shaft **25** is supported by a base (not shown) in a state in which the ring frame **21** can revolve (pivot) around a vertical central axis (pivot axis) C**2** thereof. A pivot drive mechanism (not shown) pivots the ring frame **21** around the

pivot axis C**2**.
(34) As shown in FIG. **3**, the radiation irradiation apparatus **24** is mounted on the head swing mechanism **301** having a gimbal structure, and is supported by the rotation ring **22** via the head swing mechanism **301**.
(35) When the head swing mechanism **301** does not operate, radiation Sr emitted from the radiation irradiation apparatus **24** is adjusted to pass through the isocenter C**0**, which is an intersection of the rotation central axis C**1** of the rotation ring **22** and the pivot axis C**2** of the ring frame **21**.
(36) The radiation treatment apparatus **20** further includes a sensor array **23**. The sensor array **23** receives the radiation that is emitted by the radiation irradiation apparatus **24** and that is transmitted through a subject around the isocenter C**0** to generate a transmitted image of the subject. As the sensor array **23**, an electronic portal imaging device (EPID), a flat panel detector (FPD), an X-ray image intensifier (II), or the like can be used.
(37) Further, the radiation treatment apparatus **20** includes imaging x-ray sources **26**A and **26**B and sensor arrays **27**A and **27**B. The imaging X-ray sources **26**A and **26**B and the sensor arrays **27**A and **27**B are disposed on an inner peripheral side of the rotation ring **22** and supported by the rotation ring **22**. The imaging X-ray sources **26**A and **26**B are directed to emit imaging X-rays **101** toward the isocenter C**0**. The imaging X-rays **101** are a conical cone beam. The sensor arrays **27**A and **27**B are disposed at positions facing the imaging X-ray sources **26**A and **26**B with the isocenter C**0** interposed therebetween, and receive the imaging X-rays **101** that are emitted from the imaging X-ray sources **26**A and **26**B and that are transmitted through the subject around the isocenter C**0** to generate the transmitted image of the subject. As the sensor arrays **27**A and **27**B, for example, the FPD, the X-ray II, or the like can be used.
(38) A couch drive apparatus **29** can be controlled by the control apparatus **12** to move the couch **28** in parallel with at least the rotation central axis C**1**.
(39) The head swing mechanism **301** has the gimbal structure equipped with the radiation irradiation apparatus **24**, and can tilt the radiation irradiation apparatus **24** around two shafts including a pan shaft **301***a* and a tilt shaft **301***b*. The pan shaft **301***a* is a shaft perpendicular to both the rotation central axis C**1** and the pivot axis C**2**. The tilt shaft **301***b* is a shaft parallel to the rotation central axis C**1**.
(40) As shown in FIG. **4**, the radiation irradiation apparatus **24** includes a radiation source **50**, a primary collimator **53**, a flattening filter **54**, a secondary collimator **55**, and a multi-leaf collimator (MLC) **60**.
(41) Here, the radiation source **50** is an X-ray source including an electron beam accelerator **51** and an X-ray target **52**. The electron beam accelerator **51** irradiates the X-ray target **52** with an electron beam S**0** generated by accelerating electrons. The X-ray target **52** is made of tungsten, a tungsten alloy, or the like. The X-ray target **52** emits radiation S**1** when irradiated with the electron beam S**0**.
(42) The primary collimator **53** and the secondary collimator **55** are implemented by X-ray shields (lead, tungsten, or the like) including through holes **53***h* and **55***h*, respectively, shield a part of the radiation S**1**, and emit the radiation S**1** passing through the through holes **53***h* and **55***h*.
(43) The flattening filter **54** includes a conical protrusion **54***a* made of aluminum or the like, and is disposed on an outlet side of the through hole **53***h* of the primary collimator **53**. The flattening filter **54** makes a dose distribution of the radiation S**1** uniform in a plane perpendicular to a radiation direction of the radiation S**1**.
(44) After passing through the primary collimator **53**, the flattening filter **54**, and the secondary collimator **55**, radiation S**2** having a uniform intensity distribution is incident on the multi-leaf collimator **60**. The multi-leaf collimator **60** is controlled by the control apparatus **12** to limit an irradiation field of the radiation S**2**.
(45) As shown in FIG. **7**, the multi-leaf collimator **60** has a structure in which two sets of leaf groups **70**G, in which a plurality of leaves (thin plates) **70** made of a material (lead, tungsten, or the like) that shields the radiation, are arranged in a thickness direction, face each other, and a main

plane of the leaves **70** is disposed substantially parallel to the irradiation axis of the radiation. Under control of the control apparatus **12**, a drive unit causes each leaf to protrude or retract in a direction that shields the irradiation axis of the radiation, and thus the irradiation field of the radiation S**2** can be limited or the dose distribution of the radiation in the irradiation field can be modulated.
(46) Operation of Radiation treatment System
(47) Hereinafter, in the radiation treatment system of the present embodiment, when the patient is irradiated with the radiation to perform a treatment, an operation of the treatment in an irradiation range expansion mode will be described.
(48) In the irradiation range expansion mode, the control apparatus **12** holds an irradiation axis Sc of the radiation in a state of being shifted by a predetermined amount in a predetermined direction from the isocenter C**0** by the head swing mechanism **301**, and rotates the radiation source **50** (the radiation irradiation apparatus **24**) by the rotation mechanism **302** while emitting the radiation from the radiation source with the state of the head swing mechanism maintained.
(49) First, as shown in a flow of FIG. **5**, the treatment planning apparatus **11** acquires the three-dimensional image data for the treatment plan captured in advance for the patient B (step **501**). The treatment planning apparatus **11** sets a region of interest such as a region to be irradiated with the radiation or a region to be avoided by extracting a region from the three-dimensional image by image processing, or receiving identification of the region from a user (step **502**). Next, the treatment planning apparatus **11** receives information necessary for the treatment plan such as the dose of the radiation to be emitted to each region of interest and an allowable maximum radiation dose from the user (step **503**), and creates the dose, the time, the angle, the position, the radiation region, and the like of the radiation to be radiated to the patient B as the treatment plan (step **504**). Further, an optimization calculation that takes into consideration a radiation dose rate of the radiation treatment apparatus or an operating range of each machine operating axis is performed, and an optimum intensity distribution of the radiation that can be emitted by the radiation treatment apparatus and that satisfies a radiation dose limit of each region of interest is obtained by the calculation (step **505**). For each machine operating axis at this time, each leaf drive axis of the MLC, revolving of the rotation ring **22** around the rotation central axis C**1**, pivot of the ring frame **21** around the pivot axis C**2**, and the like are selected and used.
(50) In steps **504** and **505**, the treatment planning apparatus **11** determines an amount and a direction of shifting the irradiation axis Sc of the radiation of the radiation source from the isocenter C**0** by the head swing mechanism **301**, and also calculates a parameter value of a control signal for rotating the radiation source **50** (the radiation irradiation apparatus **24**) by the rotation mechanism **302** while emitting the radiation from the radiation source **50** while maintaining the state of the head swing mechanism **301**, and a parameter value of the control signal that operates opening and closing of each leaf of the MLC **60** for implementing the optimum intensity distribution of the radiation at that time.
(51) The treatment planning apparatus **11** displays an obtained result of the treatment plan to the user (step **506**).
(52) The control apparatus **12** irradiates the patient with the radiation as follows by receiving the parameter value of the control signal calculated to implement the treatment plan by the treatment planning apparatus **11**, and controlling the head swing mechanism **301**, the rotation angle of the rotation ring **22**, and the opening and closing of the leaves of the MLC **60**.
(53) Specifically, as in step **601** of a flow of FIG. **6**, the control apparatus **12** rotates the pan shaft **301***a* and/or the tilt shaft **301***b* of the head swing mechanism **301**, sets the irradiation axis Sc of the radiation of the radiation irradiation apparatus **24** to a first irradiation direction **701** shifted by a predetermined amount from the isocenter C**0**, and, with the state maintained, rotates the radiation source **50** (the radiation irradiation apparatus **24**) by the rotation mechanism **302** by a predetermined angle range (here, 360 degrees) while emitting the radiation from the radiation

source **50** (step **601**). At this time, while the radiation source is rotated, the control apparatus **12** operates the MLC **60** to generate the intensity distribution in a radiation beam. Accordingly, the irradiation of the radiation from the radiation source **50** while rotating the radiation source **50** by the rotation mechanism **302** is referred to as rotation irradiation. That is, step **601** is first rotation irradiation.
(54) Accordingly, for example, when the first irradiation direction **701** is set on the rotation central axis C**1** by the rotation of the pan shaft **301***a* as shown in FIG. **8**A-**1**, a cylindrical irradiation range **702** is irradiated with the radiation as shown in FIG. **8**A-**2**.
(55) Next, as in step **602**, the control apparatus **12** rotates the pan shaft **301***a* and/or the tilt shaft **301***b* of the head swing mechanism **301**, sets the irradiation axis Sc of the radiation of the radiation irradiation apparatus **24** to a second irradiation direction **703** shifted by a predetermined amount from the isocenter C**0**, and, with the state maintained, rotates the radiation source **50** (the radiation irradiation apparatus **24**) by the rotation mechanism **302** by the predetermined angle range (here, 360 degrees) while emitting the radiation from the radiation source **50** (step **602**). At this time, while the radiation source is rotated, the control apparatus **12** operates the MLC **60** to generate the intensity distribution in the radiation beam. Step **602** is second rotation irradiation. At this time, a time of the rotation by the rotation mechanism can be further shortened when the direction is an opposite direction of that of the first rotation irradiation. Alternatively, in a case of a helical scan type, the rotation may be performed in the same direction as that of the first rotation irradiation by setting the irradiation axis Sc to the second irradiation direction **703** while rotating the pan shaft by a pan axis rotation mechanism, for example.
(56) Accordingly, in the first rotation irradiation and the second rotation irradiation, the irradiation axis of the radiation determined by the head swing mechanism **301** is different, and a trajectory of the rotation performed by the rotation mechanism **302** is the same. In the first rotation irradiation and the second rotation irradiation, when the trajectory is the same and the direction is opposite, the rotation is so-called revolving, but in the present specification, the rotation is used as a word including revolving for the sake of convenience.
(57) Accordingly, when the direction of shifting the second irradiation direction **703** as shown in FIG. **8**A-**1** is set to be an opposite direction of the first irradiation direction **701** (a negative direction of the rotation central axis C**1**), a cylindrical irradiation range **704** is irradiated with the radiation as shown in FIG. **8**A-**2**.
(58) The processing is repeated N times according to the treatment plan (step **603**).
(59) Accordingly, as compared with an irradiation range **705** formed when the radiation source **50** is rotated by the rotation mechanism **302** with the irradiation axis Sc of the radiation directed to the isocenter C**0** as in a comparative example, in the present embodiment, the irradiation can be performed in a total range of the irradiation ranges **702** and **704**. Therefore, the irradiation range of the radiation can be efficiently widened.
(60) Further, in the above operation of the present embodiment, if the head swing mechanism **301** holding the MLC **60** having a large weight and the radiation source **50** is once directed to the first irradiation direction **701** or the second irradiation direction **703**, the rotation may be performed by the rotation mechanism **302** while maintaining the direction, and a burden on the head swing mechanism **301** is small. Therefore, deterioration of irradiation accuracy due to a deflection of the head swing mechanism **301**, or the like is unlikely to occur, and the affected part can be irradiated with the radiation with high accuracy.
(61) Further, as shown in FIG. **8**A-**1**, the treatment plan can be set such that the irradiation range **702** of the radiation beam in the first irradiation direction **701** and the irradiation range **704** of the radiation beam in the second irradiation direction **703** (that is, an angular range of a radiation spread angle) include the isocenter C**0**, respectively. In this case, a range in which the irradiation range **702** and the irradiation range **704** overlap is generated. By generating a desired dose distribution in this overlapping range by the MLC **60**, the dose distribution of the total irradiation

range of the irradiation ranges **702** and **704** can be designed to a desired distribution.
(62) For example, as shown in FIG. **9**A, a total dose distribution in a rotation central axis C**1** direction can be flattened by designing the dose distributions in the overlapping range of the irradiation ranges **702** and **704** in a gradual decrease manner and a gradual increase manner in the rotation central axis C**1** direction. Accordingly, as in a case where the irradiation axis Sc of the radiation is aligned with the isocenter C**0** of the comparative example of FIG. **9**B, a flat dose distribution can be obtained as shown in FIG. **9**A even in a widened irradiation range.
(63) Further, as shown in FIG. **10**, by changing a degree of the gradual decrease and the gradual increase of the dose distributions in the overlapping range of the irradiation ranges **702** and **704** in the rotation central axis C**1** direction, a total dose around the isocenter C**0** can also be higher than surroundings. Accordingly, it is possible to irradiate a tumor located in the isocenter with a higher dose than the surroundings.
(64) That is, as shown in FIG. **11**, a complex dose distribution can be generated in a range **706** where the irradiation ranges **702** and **703** overlap.
Example of Expanding Irradiation Range in Rotation Radial Direction of Rotation Mechanism **302**
(65) In FIGS. **8** to **11**, an example of expanding the irradiation range in the rotation central axis C**1** direction is shown, but in the present embodiment, the irradiation range can also be expanded in a rotation radial direction.
(66) Specifically, as shown in FIG. **12**, the control apparatus **12** rotates the tilt shaft **301***b* of the head swing mechanism **301**, shifts the irradiation axis Sc of the radiation in a direction of an axis C**3** orthogonal to the rotation central axis C**1** and the pivot axis C**2**, and sets the first irradiation direction **701**. With the above state maintained, the radiation source **50** (the radiation irradiation apparatus **24**) is rotated by the rotation mechanism **302** by the predetermined angle range (here, 360 degrees) while emitting the radiation from the radiation source **50** (step **601**). At this time, while the radiation source is rotated, the control apparatus **12** operates the MLC **60** to generate the intensity distribution in the radiation beam.
(67) Accordingly, a cylindrical irradiation range **801** is irradiated with the radiation as shown in FIG. **12**. In the irradiation range **801** of FIG. **12**, a radius can be expanded to about twice that of an irradiation range **802** in the case where the irradiation axis Sc of the radiation is aligned with the isocenter C**0** as shown in FIG. **13** which is the comparative example. In addition, since the burden on the head swing mechanism **301** is small, the deterioration of the irradiation accuracy due to the deflection of the head swing mechanism **301**, or the like is unlikely to occur, and the affected part can be irradiated with the radiation with high accuracy.
Example of Expanding Irradiation Range in Rotation Axis Direction and Rotation Radial Direction of Rotation Mechanism **302**
(68) As shown in FIG. **14**, the control apparatus **12** may rotate both the pan shaft **301***a* and the tilt shaft **301***b* of the head swing mechanism **301**, shift the irradiation axis Sc of the radiation in the direction of the rotation central axis C**1** and the axis C**3**, and set the first irradiation direction **701**. With the above state maintained, the radiation source **50** (the radiation irradiation apparatus **24**) is rotated by the rotation mechanism **302** by the predetermined angle range (here, 360 degrees) while emitting the radiation from the radiation source **50** (step **601**). At this time, while the radiation source is rotated, the control apparatus **12** operates the MLC **60** to generate the intensity distribution in the radiation beam.
(69) Accordingly, a cylindrical irradiation range **803** is irradiated with the radiation as shown in FIG. **14**. In the irradiation range **803** of FIG. **14**, the irradiation range is expanded in the radial direction.
(70) Next, as shown in FIG. **15**, the control apparatus **12** rotates both the pan shaft **301***a* and the tilt shaft **301***b* of the head swing mechanism **301**, shifts the irradiation axis Sc of the radiation in the direction of the rotation central axis C**1** and the axis C**3**, and sets the second irradiation direction **703**. In the second irradiation direction **703**, the rotation direction of the pan shaft **301***a* of the head

swing mechanism **301** is an opposite direction of that in the case of FIG. **14**. With the above state maintained, the radiation source **50** (the radiation irradiation apparatus **24**) is rotated by the rotation mechanism **302** by the predetermined angle range (here, 360 degrees) while emitting the radiation from the radiation source **50** (step **602**). At this time, while the radiation source is rotated, the control apparatus **12** operates the MLC **60** to generate the intensity distribution in the radiation beam.
(71) Accordingly, as shown in FIG. **15**, the cylindrical irradiation range **804** can be expanded about twice in the radial direction and about twice in a length direction.
(72) Further, in the examples of FIGS. **12**, **14**, and **15**, the irradiation ranges **702** and **704** shifted by the head swing mechanism **301** do not include the isocenter C**0**, but as shown in FIG. **16**, it is needless to say that the irradiation range **702** can be set to include the isocenter C**0**. In this case, when the rotation mechanism **302** rotates the radiation source **50** by 360 degrees, the irradiation range **702** is doubled in the isocenter C**0** and the vicinity thereof, so that the total dose on a tumor A in the vicinity of the isocenter C**0** is larger than that in a surrounding B. Therefore, similar to the total dose distribution shown in FIG. **10**, it is possible to irradiate the tumor A located in the isocenter C**0** with a higher dose than the surrounding B.
(73) In this case, since the burden on the head swing mechanism **301** is small, the deterioration of the irradiation accuracy due to the deflection of the head swing mechanism **301**, or the like is unlikely to occur, and the affected part can be irradiated with the radiation with high accuracy.
(74) Error of Irradiation Amount and Correction Thereof
(75) In a case where the irradiation range **702** of the first rotation irradiation and the irradiation range **704** of the second rotation irradiation are partially overlapped and an entire radiation dose is made uniform, the control apparatus **12** generates the intensity distribution such that the radiation dose with respect to the overlapping portion gradually decreases and gradually increases as described above.
(76) Specifically, the control apparatus **12** operates the MLC **60** at any time during the rotation irradiation, and performs control such that an integrated amount of the radiation emitted to the overlapping portion is a desired value.
(77) However, it is difficult to control the integrated amount of the radiation with high accuracy by the control on the operation of the MLC **60** and the irradiation axis of the head swing mechanism **301**, and an error may occur.
(78) FIG. **17** is a diagram of the error of the radiation dose in the overlapping portion of the rotation irradiation. In FIG. **17**, the dose distribution is controlled such that the radiation dose of the overlapping portion gradually decreases in the first rotation irradiation (beam in the first irradiation direction), and the dose distribution is controlled such that the radiation dose of the overlapping portion gradually increases in the second rotation irradiation (beam in the second irradiation direction).
(79) However, an actual radiation dose is smaller than the dose in the treatment plan in both in the first rotation irradiation and the second rotation irradiation. Therefore, in the total dose distribution, the error is large in the region corresponding to the overlapping portion.
(80) Further, in the example of FIG. **17**, since the isocenter is included in an overlapping region and the isocenter is usually expected to be the most accurate, the error in the vicinity of the isocenter may influence the treatment plan.
(81) Therefore, in a case where the irradiation plan in which the irradiation range of the first rotation irradiation and the irradiation range of the second rotation irradiation are partially overlapped is set, the control apparatus **12** acquires the detection result by the sensor array **23** while executing at least the first rotation irradiation, corrects the radiation dose to be emitted to the overlapping portion in subsequent rotation irradiation based on the detection result, and performs the rotation irradiation reflecting the correction. In the present embodiment, the example in which the actual radiation dose is smaller than the dose in the treatment plan in the rotation irradiation is

shown, but in the case where the actual radiation dose is larger than the dose in the treatment plan, the correction can also be reflected in the next irradiation plan.
(82) FIG. **18** is a specific example of the correction of the radiation dose (No. 1).
(83) In FIG. **18**, in the first rotation irradiation (beam in the first irradiation direction), the actual radiation dose is smaller than the dose in the treatment plan.
(84) The control apparatus **12** obtains the actual radiation dose in the first rotation irradiation from the detection result obtained by the sensor array **23**, and corrects the radiation dose of the second rotation irradiation (beam in the second irradiation direction) according to the actual radiation dose.
(85) Accordingly, by acquiring the radiation dose of the first rotation irradiation and correcting the second rotation irradiation, the total dose distribution can be made uniform, and in particular, it is possible to ensure the irradiation accuracy in the vicinity of the isocenter.
(86) FIG. **19** is a specific example of the correction of the radiation dose (No. 2).
(87) In FIG. **19**, in both the first rotation irradiation and the second rotation irradiation, the actual radiation dose is smaller than the dose in the treatment plan.
(88) The control apparatus **12** obtains the actual radiation doses in the first rotation irradiation and the second rotation irradiation from the detection result obtained by the sensor array **23**, and performs additional rotation irradiation according to the total dose distribution.
(89) Accordingly, by acquiring the radiation doses of the first rotation irradiation and the second rotation irradiation and performing the additional rotation irradiation, the total dose distribution can be corrected and made uniform.
(90) The correction of the second rotation irradiation and the additional irradiation may be executed in combination.
(91) The control apparatus **12** can output the total dose distribution of a plurality of times of the rotation irradiation to the treatment planning apparatus **11**, and can reflect the total dose distribution in a next irradiation treatment. Therefore, as a result of the correction of the second rotation irradiation or the additional irradiation, even when the actual radiation dose exceeds the dose in the treatment plan, it is possible to take measures such as reducing the radiation dose in the next irradiation treatment.
(92) Processing such as the creation of the dose distribution in one rotation irradiation, the creation of the total dose distribution, the calculation of the difference between the actual radiation dose and the dose in the treatment plan may be performed with respect to the entire irradiation range, or may be performed selectively for the overlapping portion. A processing load can be reduced by selectively performing the processing on the overlapping portion and reflecting the processing in the subsequent rotation irradiation.
(93) FIG. **20** is a flowchart showing a processing procedure for the correction of the error. In the processing procedure shown in FIG. **20**, the control apparatus **12** executes the correction of the second rotation irradiation and the additional irradiation in combination. Specifically, the control apparatus **12** first executes a first time of rotation irradiation (first rotation irradiation) in the first irradiation direction (step **2001**). Then, the control apparatus **12** acquires the dose distribution of the first rotation irradiation from the detection result obtained by the sensor array **23** (step **2002**).
(94) The control apparatus **12** compares the dose distribution of the first rotation irradiation with that in the treatment plan, and corrects the dose of a second time of the rotation irradiation (second rotation irradiation) (step **2003**).
(95) The control apparatus **12** executes the second time of rotation irradiation in the second irradiation direction reflecting the correction (step **2004**). Then, the control apparatus **12** acquires the dose distribution of the second rotation irradiation from the detection result obtained by the sensor array **23** (step **2005**).
(96) The control apparatus **12** obtains the total dose distribution, compares the total dose distribution with that in the treatment plan, and determines whether the additional irradiation is necessary (step **2006**).

(97) If the additional irradiation is necessary (step **2006**; Yes), the control apparatus **12** executes the additional rotation irradiation (step **2007**), and acquires the dose distribution of the additional rotation irradiation (step **2008**). The control apparatus **12** updates the total dose distribution using the acquired additional dose distribution.
(98) After a completion of step **2008**, or when the additional irradiation is not necessary (step **2006**; No), the control apparatus **12** outputs the total dose distribution (step **2009**) and ends the processing.
(99) As a modification, the irradiation may be controlled with reference to a long-term treatment plan.
(100) FIG. **21** is a diagram of irradiation control according to the treatment plan. In FIG. **12**, if there is a schedule of the next irradiation treatment within a predetermined number of days, the irradiation control is adopted in which "A second time of radiation dose is corrected based on a first time of dose distribution, and a total of the two doses is adjusted to a target dose.". On the other hand, when there is no schedule of the next irradiation treatment within the predetermined number of days, the irradiation control is adopted in which "The first and second times of radiation doses are set to be small, and the radiation doses are adjusted to the target dose in the additional irradiation.".
(101) Accordingly, if the control apparatus **12** adopts the irradiation control according to the treatment plan, the irradiation can be reliably ended with the time required for the two times of rotation irradiation when the excess or deficiency can be corrected in the next irradiation treatment, and careful irradiation can be performed on a premise of the additional irradiation when the excess or deficiency cannot be expected to be corrected in the next irradiation treatment.
(102) In the above description, the case where the irradiation range is widened by partially overlapping the irradiation ranges of the two times of rotation irradiation has been described as an example, but the irradiation range can be further widened by using three or more times of the rotation irradiation.
(103) FIG. **22** is a diagram showing widening of the irradiation range using three times of the rotation irradiation.
(104) In FIG. **22**, an irradiation range **2202** (of the second rotation irradiation) of the beam in the second irradiation direction includes the isocenter C**0** in the vicinity of a center. Then, an irradiation range **2201** (of the first rotation irradiation) of the beam in the first irradiation direction partially overlaps with the irradiation range **2202**. Further, an irradiation range **2203** (of third rotation irradiation) of the beam in a third irradiation direction is an opposite direction of that of the irradiation range **2201** with the isocenter C**0** as a center, and partially overlaps with the irradiation range **2202**.
(105) When the first to third rotation irradiation is performed in this manner, the total dose distribution can also be made uniform by controlling the dose distribution in each overlapping portion.
(106) Modification Applied to Non-rotation Irradiation
(107) In the above description, the case where the irradiation ranges of the first rotation irradiation and the second rotation irradiation are partially overlapped has been described as an example, but the invention is not limited thereto. For example, first irradiation and second irradiation may be performed in which the irradiation ranges are partially overlapped by fixing the rotation mechanism **302** and controlling the head swing mechanism **301**.
(108) Accordingly, as well in non-rotation irradiation, which is the irradiation with the rotation mechanism **302** fixed, the detection result obtained by the sensor array **23** can be acquired, and the radiation dose emitted to the overlapping portion in subsequent irradiation can be corrected based on the detection result.
(109) In the non-rotation irradiation, the radiation dose can also be adjusted by operating the multi-leaf collimator **60** during irradiation. Specifically, by gradually moving the leaves for the

overlapping portion, the gradual decrease and the gradual increase of the radiation dose are implemented.
(110) In the rotation irradiation, since the head swing mechanism **301** or the like having a weight is rotated by the rotation mechanism **302**, a mechanical error of hardware occurs. On the other hand, in the non-rotation irradiation, by performing the irradiation with the rotation mechanism **302** fixed, an error in an entire irradiation amount can be reduced. Further, by using the control of the multi-leaf collimator **60**, highly accurate irradiation can be ensured in the overlapping portion where the error is large.
(111) As described above, the disclosed radiation treatment system includes: the couch **28** that carries the treatment target, the radiation source **50**, the rotation mechanism **302** that supports the radiation source **50** and that rotates the radiation source **50** around the couch **28**, the sensor array **23** as the sensor that detects the radiation transmitted through the treatment target, and the control apparatus **12** as the control unit that controls the radiation source **50** and the rotation mechanism **302**, and the control unit sets an irradiation plan in which an irradiation range of first irradiation and an irradiation range of second irradiation are partially overlapped, and controls a radiation dose for an overlapping portion based on a detection result obtained by the sensor.
(112) Therefore, the disclosed radiation treatment system enables wide range irradiation with high accuracy.
(113) According to the disclosed radiation treatment system, the control unit is able to execute rotation irradiation in which the radiation is emitted from the radiation source while rotating the radiation source by the rotation mechanism, and the first irradiation and the second irradiation are the rotation irradiations, respectively.
(114) Therefore, the irradiation range can be expanded by partially overlapping the irradiation ranges in the rotation irradiation, and the radiation dose in the expanded irradiation range can be controlled with high accuracy.
(115) The disclosed radiation treatment system further includes the head swing mechanism **301** that swings the radiation source so as to swing an irradiation axis of the radiation, in the rotation irradiation, the radiation is emitted from the radiation source **50** while rotating the radiation source **50** by the rotation mechanism **302** with the state of the head swing mechanism **301** maintained, and the first irradiation and the second irradiation are the rotation irradiations in which irradiation axes of the radiation are different and trajectories of the rotation performed by the rotation mechanism are the same.
(116) Therefore, it is possible to avoid a situation in which the accuracy of the radiation irradiation is reduced due to operation accuracy of the head swing mechanism **301**.
(117) The control unit can acquire an irradiation result of the first irradiation by the sensor and can correct a radiation dose to be emitted to the overlapping portion in the second irradiation.
(118) In addition, the control unit can acquire the irradiation results of the first irradiation and the second irradiation by the sensor, and can perform the additional irradiation when the radiation dose with respect to the overlapping portion is insufficient.
(119) Accordingly, the disclosed radiation treatment system can correct the error in the radiation dose by the correction of the second irradiation or the additional irradiation.
(120) The disclosed radiation treatment system further includes the multi-leaf collimator **60** that partially shields the radiation so as to form a shape of the irradiation range of the radiation, and the control unit controls an integrated amount of the radiation emitted to the overlapping portion by operating the multi-leaf collimator at any time while performing the irradiation from the radiation source.
(121) Therefore, the intensity distribution can be generated in the radiation beam, which contributes to the highly accurate irradiation.
(122) The control unit can selectively obtain a difference between a radiation dose in the irradiation plan and a radiation dose actually emitted to the overlapping portion, and reflect the difference in

subsequent irradiation.
(123) According to this control, it is possible to implement the highly accurate irradiation while preventing a load on the calculation.
(124) The control unit can output a total dose distribution obtained based on a plurality of times of irradiation and reflect the total dose distribution in a next irradiation treatment.
(125) The control unit can determine the irradiation plan according to a time until a next irradiation treatment in a treatment plan.
(126) Therefore, an adaptive radiation treatment can be performed in conjunction with the treatment plan.
(127) The invention is not limited to the above disclosed contents, and includes various modifications. For example, the embodiment described above has been described in detail for easy understanding of the invention, and the invention is not necessarily limited to those including all the configurations described above. In addition, the configuration is not limited to being deleted, and the configuration may be replaced or added.
(128) For example, in the above embodiment, the irradiation range of the first rotation irradiation and the irradiation range of the second rotation irradiation are partially overlapped by using the head swing mechanism **301**, but the irradiation range of the first rotation irradiation and the irradiation range of the second rotation irradiation may be partially overlapped by moving the couch.
(129) The irradiation range of the first rotation irradiation and the irradiation range of the second rotation irradiation may be partially overlapped by using not only the head swing mechanism **301** but also another mechanism such as a slide mechanism.

## Claims

1. A radiation treatment system comprising: a couch that carries a treatment target; a radiation source; a rotation mechanism configured to support the radiation source and to rotate the radiation source around the couch; a sensor configured to detect radiation transmitted through the treatment target; and a control unit configured to control the radiation source and the rotation mechanism, a head swing mechanism configured to swing the radiation source so as to swing an irradiation axis of the radiation; and a multi-leaf collimator configured to partially shield the radiation so as to form a shape of an irradiation range of the radiation, wherein the control unit is configured to execute rotation irradiation in which the radiation is emitted from the radiation source while the radiation source is rotated by the rotation mechanism, wherein a first irradiation and a second irradiation are the rotation irradiation in which irradiation axes of the radiation are different and trajectories of the rotation performed by the rotation mechanism are the same, wherein the control unit is configured to: set an irradiation plan in which an irradiation range of the first irradiation and an irradiation range of the second irradiation partially overlap, control an integrated amount of the radiation emitted to the overlapping portion by causing the multi-leaf collimator to operate at any time while performing the irradiation from the radiation source, and selectively obtain a difference between a radiation dose in the irradiation plan and a radiation dose actually emitted to the overlapping portion based on a detection result obtained by the sensor and reflect the difference in a subsequent irradiation.
2. The radiation treatment system according to claim 1, wherein in the rotation irradiation, the radiation is emitted from the radiation source while rotating the radiation source by the rotation mechanism with a state of the head swing mechanism maintained.
3. The radiation treatment system according to claim 1, wherein the control unit is configured to acquire an irradiation result of the first irradiation by the sensor and corrects the radiation dose emitted to the overlapped portion in the second irradiation.
4. The radiation treatment system according to claim 1, wherein the control unit is configured to

acquire irradiation results of the first irradiation and the second irradiation by the sensor, and perform additional irradiation when the radiation dose with respect to the overlapped portion is insufficient.

5. The radiation treatment system according to claim 1, wherein the control unit is configured to outputs a total dose distribution obtained based on a plurality of times of irradiation and reflects the total dose distribution in a next irradiation treatment.

6. The radiation treatment system according to claim 1, wherein the control unit is configured to determine the irradiation plan according to a time until a next irradiation treatment in a treatment plan.

7. A method of operating a radiation treatment system including: a couch that carries a treatment target, a radiation source, a rotation mechanism configured to support the radiation source and to rotate the radiation source around the couch, a sensor configured to detect radiation transmitted through the treatment target, a head swing mechanism configured to swing the radiation source so as to swing an irradiation axis of the radiation, a multi-leaf collimator configured to partially shield the radiation so as to form a shape of an irradiation range of the radiation, a control unit configured to control the radiation source and the rotation mechanism, the method comprising steps executed by the control unit: executing rotation irradiation in which the radiation is emitted from the radiation source while the radiation source is rotated by the rotation mechanism, wherein a first irradiation and a second irradiation are the rotation irradiation in which irradiation axes of the radiation are different and trajectories of the rotation performed by the rotation mechanism are the same; setting an irradiation plan in which an irradiation range of the first irradiation and an irradiation range of the second irradiation partially overlap; controlling an integrated amount of the radiation emitted to the overlapping portion by causing the multi-leaf collimator to operate at any time while performing the irradiation from the radiation source; and selectively obtaining a difference between a radiation dose in the irradiation plan and a radiation dose actually emitted to the overlapping portion based on a detection result obtained by the sensor and reflect the difference in a subsequent irradiation.